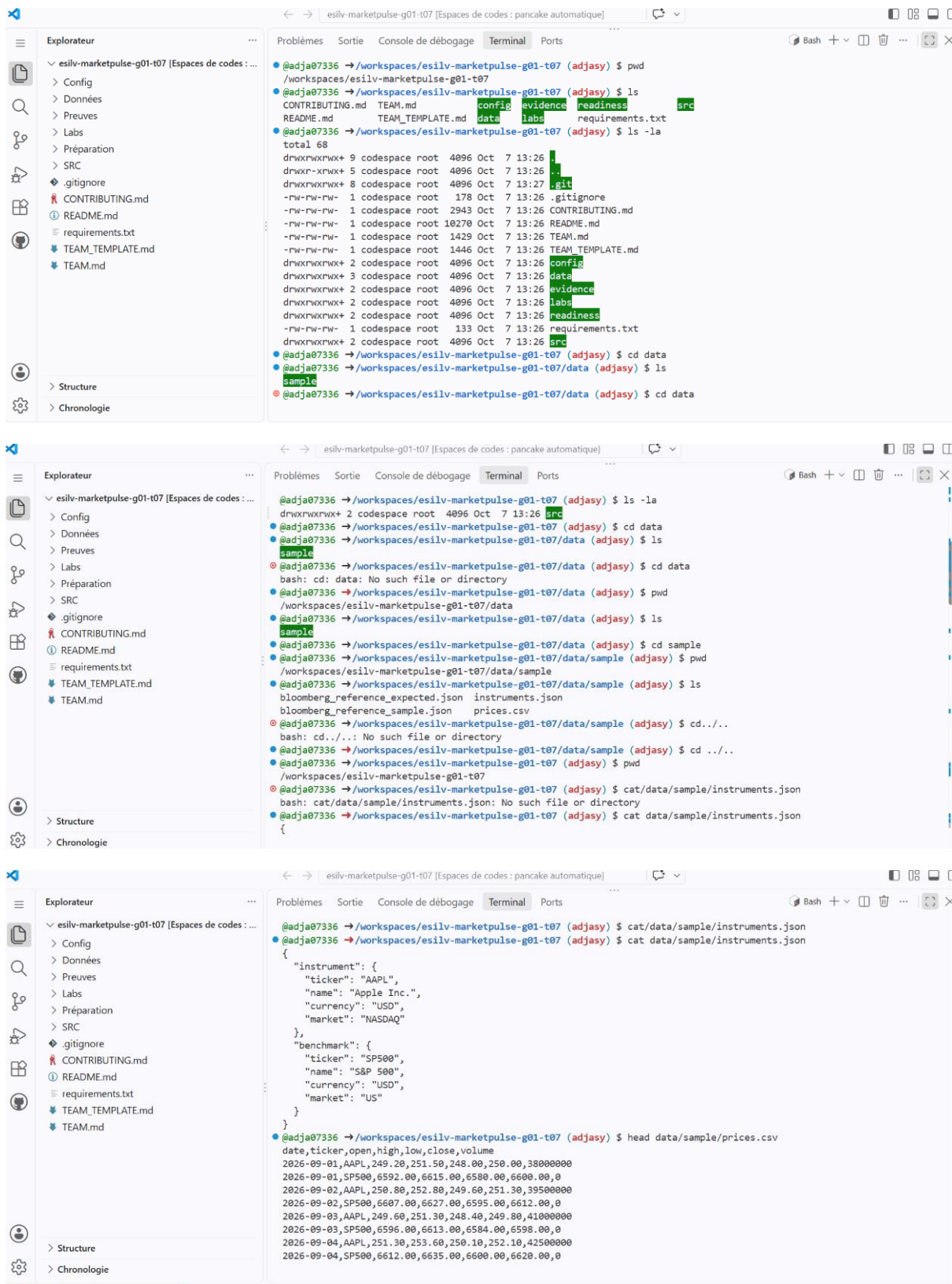


## TD01\_BOOSTRAP\_LINUX:



The image displays three sequential screenshots of a Visual Studio Code editor window, showing the file explorer on the left and the terminal output on the right. The project is named 'esilv-marketpulse-g01-t07' and is located in the 'Espaces de codes : pancake automatique' workspace.

**First Screenshot:** The file explorer shows the project structure, including files like 'Config', 'Données', 'Preuves', 'Labs', 'Préparation', 'SRC', '.gitignore', 'CONTRIBUTING.md', 'README.md', 'requirements.txt', 'TEAM\_TEMPLATE.md', and 'TEAM.md'. The terminal shows the following commands and output:

```
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ pwd
/workspaces/esilv-marketpulse-g01-t07
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ ls
CONTRIBUTING.md  TEAM.md
README.md        TEAM_TEMPLATE.md  data  labs  requirements.txt
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ ls -la
total 68
drwxrwxrwx+ 9 codespace root 4096 Oct 7 13:26 .
drwxr-xrwx+ 5 codespace root 4096 Oct 7 13:26 ..
drwxrwxrwx+ 8 codespace root 4096 Oct 7 13:27 .git
-rw-rw-rw- 1 codespace root 178 Oct 7 13:26 .gitignore
-rw-rw-rw- 1 codespace root 2943 Oct 7 13:26 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 10270 Oct 7 13:26 README.md
-rw-rw-rw- 1 codespace root 1429 Oct 7 13:26 TEAM.md
-rw-rw-rw- 1 codespace root 1446 Oct 7 13:26 TEAM_TEMPLATE.md
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:26 config
drwxrwxrwx+ 3 codespace root 4096 Oct 7 13:26 data
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:26 evidence
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:26 labs
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:26 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 13:26 requirements.txt
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:26 src
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ cd data
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ ls
sample
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ cd data
```

**Second Screenshot:** The terminal shows the following commands and output:

```
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ ls -la
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:26 src
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ cd data
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ ls
sample
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ cd data
bash: cd: data: No such file or directory
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ ls
sample
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data (adjas) $ cd sample
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjas) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data/sample
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjas) $ ls
bloomberg_reference_expected.json  instruments.json
bloomberg_reference_sample.json    prices.csv
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjas) $ cd ../
bash: cd: ../: No such file or directory
@adja07336 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjas) $ cd ../
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ pwd
/workspaces/esilv-marketpulse-g01-t07
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ cat data/sample/instruments.json
bash: cat/data/sample/instruments.json: No such file or directory
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ cat data/sample/instruments.json
{
```

**Third Screenshot:** The terminal shows the following commands and output:

```
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ cat data/sample/instruments.json
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
```

1. The instrument is:

Ticker: AAPL

Name: Apple Inc.

Currency: USD  
Market: NASDAQ

2. The benchmark is:

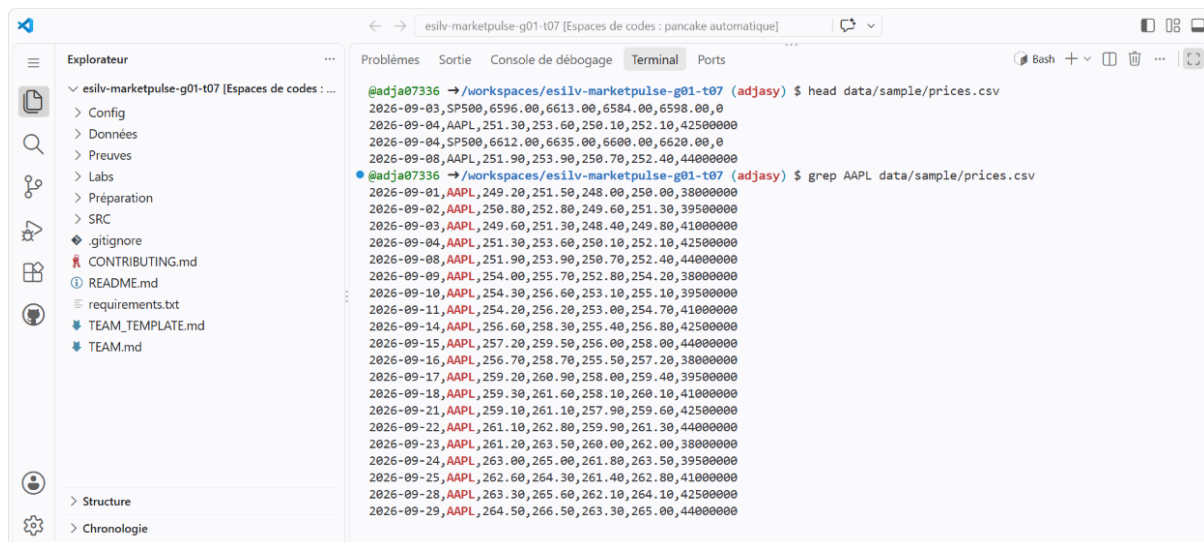
Ticker: SP500

Name: S&P 500

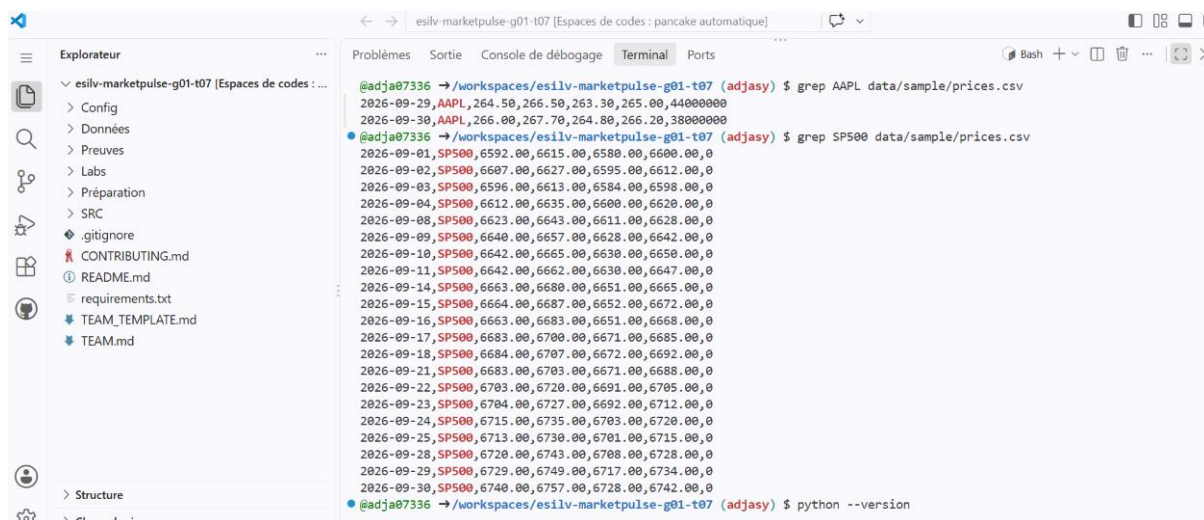
Currency: USD

Market: US

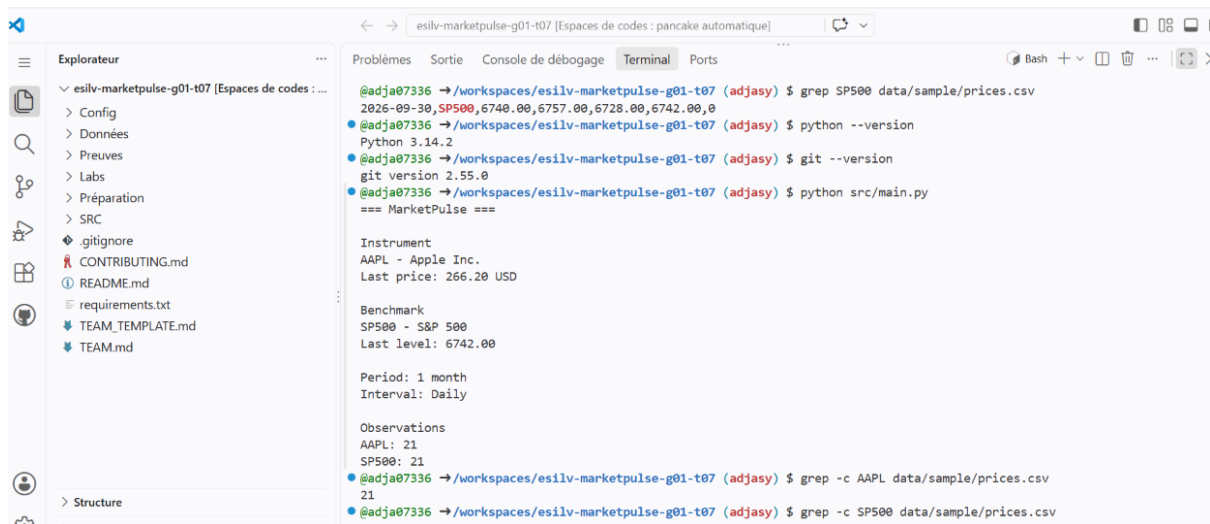
3. Their names and tickers are stored in the instruments.json file:  
data/sample/instruments.json



```
esilv-marketpulse-g01-t07 [Espaces de codes : pancake automatique]
Problèmes Sortie Console de débogage Terminal Ports
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ head data/sample/prices.csv
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep AAPL data/sample/prices.csv
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
```



```
esilv-marketpulse-g01-t07 [Espaces de codes : pancake automatique]
Problèmes Sortie Console de débogage Terminal Ports
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep AAPL data/sample/prices.csv
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep SP500 data/sample/prices.csv
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
@adja07336 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ python --version
```



```
@adja07336 →/workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep SP500 data/sample/prices.csv
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
@adja07336 →/workspaces/esilv-marketpulse-g01-t07 (adjas) $ python --version
Python 3.14.2
@adja07336 →/workspaces/esilv-marketpulse-g01-t07 (adjas) $ git --version
git version 2.55.0
@adja07336 →/workspaces/esilv-marketpulse-g01-t07 (adjas) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
@adja07336 →/workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep -c AAPL data/sample/prices.csv
21
@adja07336 →/workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep -c SP500 data/sample/prices.csv
```

## TD02 PYTHON CSV JSON:

1. The function `load_instruments()` reads JSON
2. The function `load.prices ()` reads csv
3. The function `filter_prices(prices, ticker)` selects one sticker
4. The function `main ()` coordinates execution
5. They are stored in `data/sample`